

## CHEM 101 Topic 11: Acids and Bases — Problem Set

*Instructions: Complete the following problems using the analysis method practiced in your guided assignment. For pH calculations, show your log formula. For net ionic equations, provide the full ionic equation first, identify the spectator ions, and then write the final net ionic equation.*

### Section 1: Defining Acids and Bases

Problem 1: Is NaOH an Arrhenius acid or an Arrhenius base? Justify your answer.

Problem 2: In the reaction  $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ , identify the Bronsted-Lowry acid and the Bronsted-Lowry base.

Problem 3: Explain why a substance that is an Arrhenius acid must also be a Bronsted-Lowry acid, but a Bronsted-Lowry acid is not necessarily an Arrhenius acid.

### Section 2: Water, Ionization, and pH

Problem 4: A solution has a pH of 8. Is it acidic, basic, or neutral?

Problem 5: Calculate the pH of a solution where  $[\text{H}^+] = 2.5 \times 10^{-3} \text{ M}$ .

Problem 6: Solution A has a pH of 3.0, and Solution B has a pH of 5.0. How many times more concentrated in  $\text{H}^+$  is Solution A than Solution B?

### Section 3: Electrolytes & Net Ionic Equations

Problem 7: Classify the following as a strong electrolyte or a weak electrolyte:  $\text{HNO}_3$  (nitric acid) and  $\text{HF}$  (hydrofluoric acid).

Problem 8: In the reaction  $\text{KNO}_3(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{KCl}(\text{aq}) + \text{HNO}_3(\text{aq})$ , identify the spectator ions.

Problem 9: Write the net ionic equation for the reaction between aqueous barium nitrate,  $\text{Ba}(\text{NO}_3)_2$ , and aqueous sodium sulfate,  $\text{Na}_2\text{SO}_4$ .

### Section 4: Clinical Applications

Problem 10: A patient's blood pH is measured at 7.30. Is this an example of acidosis or alkalosis?

Problem 11: Using the concept of neutralization, explain how a base-based antacid treats the symptoms of an overproduction of  $\text{HCl}$  in the stomach.

Problem 12: Sodium bicarbonate ( $\text{NaHCO}_3$ ) is used in medical settings to treat metabolic acidosis. Based on its role as a strong electrolyte, explain how it helps raise blood pH.